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Conference Paper · October 2023

DOI: 10.1109/EASCT59475.2023.10392472

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College Bus Number Plate Registration Detection Using YOLO-V8

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Abstract*—There is a need for automation of vehicle entering system and to reduce the time of verification. We propose a solution for this problem considering college buses o BVRIT as organization. College bus number plate Registration Detection is crucial part of smart BVRIT planning and BVRIT transport management. This paper presents YOLO V8 algorithm for Detecting college bus using number plate registration detection. YOLO - You Only Look Once. This YOLO-V8 algorithm is state-of-the-art (SOA) model or algorithm. It's real-time object detection algorithm that can detect objects in any image, that has many classes in it with high accuracy and speed. A dataset of College bus images is collected manually and finetuned using transfer learning. Model is trained on powerful GPU (Graphics Processing Unit), significantly reducing training time and taking less time for Character recognition. The model's accuracy is computed by average, precision, recall, and F1 score as well curves. Our results show that Yolo V8, model can accurately detect the bus with high accuracy and speed in any weather condition. The text from detected number plate is recognized using Easy-OCR. This number plate text is used to infer whether bus is of COLLEGE BUS or NON-COLLEGE BUS. Model's overall performance is remarkable. While maintaining recall of 95 percent, which improves its capacity to collect a substantial number of pertinent things, it attained a precision of 99.2 percent, suggesting a high accuracy in properly identifying objects. The model's ability to detect items with a high degree of confidence is demonstrated by its mAP50 score of 99 Percent. This is one of the optimal solutions to minimize human work and time, making automation. This research has been done with the purpose to find the registered college bus is supposed to enter the college campus and Non-college bus is not supposed to enter the campus.

Keywords—YOLO V8, Easy-OCR, Character Recognition, Object detection, Recall, Precision, mAP(mean Average precision), License plate detection, Bounding box.

I. Introduction

In general, at many companies, the vehicles form a queue to go inside due to paperwork that deals with verification of a vehicle, It checks whether it belongs to the organization or an external vehicle. This consumes a large amount of time. There is a need for automation of this system and to reduce time of verification. In this research, we present an efficient way to minimize the time of vehicle verification. As we cannot get real time data on the vehicles as it is very confidential data, In this research we use BVRIT (B V RAJU INSTITUTE OF TECHNOLOGY) as organization and buses as vehicles. BV Raju Institute of Technology is a prestigious engineering college located in Telangana, India. There is no proper system for monitoring the buses whether they belong to BVRIT or not. In this paper, a real-time object detection algorithm – YOLO-V8 for detecting the bus and number plate. The bounding boxes of the number plate are extracted and in turn, the Alpha Numeric-characters like Number plate digits and the Alphabet of the number plate are Recognized using the Optical Character Recognition (OCR) technique and are extracted and checked whether the bus belongs to college (BVRIT) or any Non-college bus. YOLO(You Only Look Once) is an algorithm that can detect even very small objects in the image in weather conditions like low lighting. This makes it useful for our research. YOLO algorithms enable real-time detection of the classes which have to detect along with the prediction of the bounding boxes. Yolo algorithms show the class detection probabilities. YOLO algorithms enable real-time object detection by simultaneously predicting the bounding box coordinates and class probabilities in a single pass through the Convolutional neural networks which are present in the Architecture of the YOLO. Due to the addition of modules for spatial attention, feature fusion, and context aggregation, the most recent version of the YOLO algorithm, known as YOLOv8, performs better than earlier versions. These improvements have made YOLOv8 one of the most significant object-detecting systems in the business. The model's performance (ability to detect) is evaluated using standard evaluation metrics including recall, F1 score, precision, and average score. Using LabelImg, the 45 images of the dataset from College transport bus were manually classified into two categories: train (35 images of college buses(bvrit) buses) and Val (8 images of bvrit buses) and (5 images of bvrit buses) for the test. After all, the model can detect the Bus and number plate present in the College bus image in one single pass with improved accuracy. And next, the text extraction is done and it is checked in the file containing the data of the number plates and produces the output as COLLEGE-BUS or NON-COLLEGE BUS. The number plate is the vehicle's distinctive identification, and it is used as evidence of the legitimacy of the vehicle's operation. It is issued by the police and is mounted on the vehicle and contains the area code, registration number, and validity period[1].

II. Related Work

The 3 stages for detection of the Bus and number plate are, first comes license plate detection and next character segmentation, and After that character identification from the

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number plate. License plate detection is done to find the exact location of the license plate from the image. Character segmentation deals with the separation and filtration of characters that are present on the license plate that has been detected using any algorithm. Characters that have been separated and filtered using the preprocessing techniques are then identified so that the letters and numbers present on the license plate are known[1]. This can be done using image or video surveillance cameras[2]. A method for vehicle counting and number plate recognition utilizing YOLOv3 was proposed by Gomathy. In 2021, There research study detected car number plates and the accuracy with which they detected is 95 percent. They used the Indian vehicle as their dataset in the research. Their model could be used for the detection of the vehicle and number plate, recognizing the text present in the number plate, tracking the vehicles in the video, and then classifying the detected vehicles according to car, bus, etc... to their size into three different classes. The proposed system contains 3 divisions, they are 1. background learning, 2. foreground extraction, and 3. vehicle classification. Using background subtraction we can extract the foreground image and detect the objects in the image[3]. The number plates can have different languages they can be extracted using OCR techniques like "Pytessrect", and "EasyOCR" [4]. A research study article on Single Shot MultiBox Detector - SSD was published by an author named W. Liu et al in the year 2016. This SSD uses the base network "VGG16" for high image-quality classification. To predict the objects at different scales and aspect ratios the size of convolutional feature layers are reduced and added to the end of the VGG16.[5]. This vehicle detection gave scope for vehicle information retrieval which helped the government to track the vehicle's fines[6].To Recognise the characters in the number plate the image must undergo some image preprocessing techniques like grayscale conversion, image thresholding getting the contours applying the filters and, masks for noise reduction in images [7]. The architecture of the YOLO algorithm essentially consists of three levels. The remainder block, the detection layer, and the upper sampling layer are the three essential layers in the YOLO. Oversampling improves network learning of fine grained features, which is critical for detecting tiny objects [8]. The updated model of the YOLOv5 is yolo v7 which is very efficient in detecting objects. LPRNet is abbreviated as a License plate Recognition network used for recognition of the characters and extracting them from the image of the number. It is of the techniques in license plate recognition. It is a deep learning model trained above the task.[9]. Using data from automatic number plate recognition cameras, Hadavi et al. (2020) examined the movements of both passenger and freight vehicles. They acquired a vehicle detection accuracy of 97.3 percent using YOLOv3 [10]. Using improved network structure, targeted downsampling, and feature fusion, the study introduces the ground-breaking DC-YOLOv8 algorithm, which beats existing models for small size target recognition in accuracy[11]. The paper demonstrates the enhanced object detection capability, speed, and application of YOLOv5 in a range of different real-world scenarios[12]. Unmanned surface vehicles are empowered by YOLOv5-based lightweight object detection, which increases speed and accuracy in aquatic conditions[13]. The paper reviews how YOLO models evolved for the real time industrial surface defect detection on edge devices[14]. Introducing YOLO-GNS, a cutting-edge method that maximises precision and computing efficiency for UAV-based special vehicle detection[15].

THE CONTRIBUTION OF OUR RESEARCH IS :

1. Accuracy of YOLO-v8 in detecting college bus is well known. It uses in college bus number plate registration can lead to further precise number plate recognition, specifically in difficult low lightening circumstances.
2. Implementing YOLO v8 for college bus recognition will enable authorities to identify and track vehicles more effectively.
3. The updated version built on yolov5,v7 is Yolo v8.YOLO v8 is the current version of the YOLO. By advancing preceding versions numerous alterations like 1. Spatial attention 2. Feature fusion and 3. context aggregation are brought in. From these advances, YOLO v8 can precisely detect objects faster with more accuracy. These all make YOLO v8 one of the most significant in the field of object detection. This gives us the research scope for using the YOLO V8 algorithm.

III. Problem statement

The existing methods at the companies at entry and exit gates use paperwork for the verification of vehicles costing most of the time. There is no proper verification system that can minimize the time of verification. This large amount of time leads to traffic

- To identify whether the bus belongs to college BVRIT(organization) using YOLO v8.
- There is a need for automation of this verification and to minimize the time of verification.
- Current Existing methods which use deep learning and the preceding Yolo algorithm are not accurate in the detection of vehicles when they are in different scales or weather conditions and they are facing challenges in detecting very small and very large objects in low lighting conditions.

IV. Objectives

1) The goal of the research is to minimize the verification time at entry and exit points in the organization.
2) To conserve the time at entry and exit points avoiding the large queue of vehicles and traffic in the organization.
3) Detect whether the bus is a BVRIT or NON-COLLEGE BUS bus.
4) To develop an Object detection algorithm that has highspeed detection for small to large objects in any weather conditions.
5) yolo v8 (Yolov8x.pt) model runs on GPU which reduces the model training time and increases the accuracy of detection and can detect any object.
6) Yolo v7 and v8 are almost similar in accuracy but differ In speed and processing time this can be achieved by Yolo v8 by changing the runtime from CPU to GPU and using a new anchor-free detection head.

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V. Existing Work

To detect vehicles but accuracy of detection is less compared to our model which is previous models. yolov7 is similar to yolo-v8 accuracy of detection became same for yolo-v8 and yolov7. The existing works of vehicle detection are based on the models, they are yolo v1 to yolo v7 and now v8. Many pieces of research have been conducted for object detection from Yolo v1 to yolov8. Research study on Indonesian License Plate Detection and Identification Using Deep Learning showed an accuracy of 95 percent [1].ANPR Indian system uses Surveillance Cameras car number plate detection rate of 98.5 percent using Support Vector Machine algorithm [2]. Gomathy, C. K., Sandeep Sai, A. Pradeep Reddy, and C. Nitish(2021). Number plate recognition and vehicle count. This work proposes a mechanism for counting vehicles and identifying license plates. Accuracy was 94 percent because of OpenCV library, which authors used [3]. Plate Number Recognition based on Hybrid Techniques with 92.2 percent accuracy [4]. Vehicle Information Retrieval by Number Plate Detection, After testing the model, got a 92 percent of accuracy which is more than any other machine learning algorithms like KNN, and Naive Bayes. G, Kavya, S.Manjushree, V Ranjitha, and E Mamatha in the year 2020 published a research paper titled- Number Plate Detection for Recovery of Vehicle Information[6].Sweta Khandekar (2021), SayaliLokhande, Pragati Sahane, Sonali Katare, and Dnyaneshawari Tawhare. Utilizing machine learning, an automatic number plate recognition system can identify vehicles. The accuracy was 98 percent and authors used a dataset of 1000 photos [7]. A research study on "License Plate Detection and Recognition Systems based on YOLOv7" using LPRNet, was able to detect the vehicle number plate at any weather conditions like rainy, low-level light conditions with a Recall of 95.17 percent and 95.13 percentage, and Precision of 97.77 percentage and 97.79 percentage[8]. Huang, H., Macharis, C, Ms.Hadav, S.Rai, H. B.Verlinde, and Guns, T. in the year 2020 had done a research study examining the data from automatic number plate recognition cameras to determine the motion in humans and vehicles.The authors obtained an accuracy of 85 percent using data from ANPR cameras in Belgium [9].

VI. Proposed work

To minimize the time of verification at entry and exit points we propose an optimal solution that uses the YOLO algorithm and version V8 which is the up-to-date version of the YOLO. The solution has three steps to follow which are described below. The constraint for the research is the data of the buses of an organization must be filed or stored in a database. When the vehicle enters the gate the camera at the gate automatically detects the vehicle and verification is done in an automated manner, permission is granted for entry, and gates open for vehicle entry, or else the message is displayed as no entry and the vehicle must not enter. This reduces the time for verification and labour work and the paperwork at entry and exit points.In previous paper methods used are YOLO v5,YOLO v7.but in this research work we have done the novelty of YOLO v8 version, we are performing this yolo v8 in this research using yolo v8 is to detect with high accuracy & high speed. Whereas YOLO v5,yolo v7, there accuracy will be less compared to YOLO V8 in low lightening conditions.The major feature of YOLO is the single-stage detection technique, which is well-intentioned to detect objects in real world and with improvement in accuracy. In a single pass, the YOLO can detect the objects present in the image, making it faster and more effective than two-stage detection methods like R-CNN As they have two-stage detection first extract regions of interest and then classify these regions. Yolo is faster than R-CNN This section describes in detail the proposed approach of "College bus Detecting and license plate registration Recognition using the Yolo v8", divided into 3 parts:

1. Bus and License Plate Detection.
2. Extracting the region of interest and preprocessing.
3. Recognizing the Characters on License Plate.

A. Bus and License Plate Detection

Bus and License Plate Detection In the very first step, the bus image is captured using a mobile and then the image is given to the custom YOLO v8 model for Bus and License plate detection. The GPU-centric, efficient algorithm we are proposing to use to detect the object area from the image is the YOLO(You Only Look Once) V8 algorithm. What is the YOLO algorithm? When YOLO came into the picture of Deep learning Object Detection algorithms, Many algorithms were proposed to detect the object from the image but YOLO took a complexly different approach. The fact is that it actually looked once but in a clever way. The first version of YOLO was proposed in 2016 and the latest version YOLO v8 is proposed on 10th Jan 2023 which we are going t talk about in this research paper. YOLO object detector is faster than traditional object detectors, as it is trained on COCO (Common Objects in Context) dataset and it belongs to the family of one-stage detectors which makes it faster. These object detectors need to be run on GPU which requires more time.YOLO v8 proposed this issue by making object detection possible using a single GPU and reducing the training time as YOLOv8 uses Anchor free detection. YOLO can detect objects under any weather condition and lighting.

How does YOLO work? YOLO first divides the image in 13*13 of which each cell detects five bounding boxes (the bounding box is the rectangle that encloses the object). YOLO gives a confidence score which shows how confident the algorithm is that the object to detect exists in the bounding box. The score doesn't tell the type of object but tells the confidence score if it is in that bounding box or not. Now we know there are n*n grid cells and each detects 5 bounding boxes, so the total bounding boxes will be n*n*5 which is 185. For example, there are 13*13 grid cells and each cell detects 5 bounding boxes so the total bounding boxes become 13*13*5 which makes 845 bounding boxes. Now the beauty of the algorithm is that these total 845 boxes were predicted all at once as the name suggests "You Only Look Once".

How YOLO is better than R-CNN? R-CNN takes a huge amount of time to train the model, however, YOLO takes comparatively very less amount of time to train the same model. R-CNN(Region Convolutional Neural Network can't be used for real-time implementation of object detection as it takes 47 seconds for each test image but YOLO can be used in real-time very efficiently.

B. Extracting the region of interest and preprocessing

Extraction of License Plate The very second step after the detection is the extraction of the License Plate area, That is Region of interest extracted from the detected Image. The process is as follows :

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1) Extraction of Bounding box coordinates: The bounding box coordinates extraction is done based on results obtained from the custom YOLO V8 model. The coordinate of the bounding boxes is obtained as x1, x2, x3, and x4. The bounding box coordinates xc, yc, width, and height are extracted using as shown in the equation ((1),(2),(3),(4)) as shown below.

$$x_c = \frac{x_1+x_2}{2} \quad (1)$$

$$y_c = \frac{x_3+x_4}{2} \quad (2)$$

$$width = x2 - x1 \quad (3)$$

$$height = y2 - y1 \quad (4)$$

2) Extraction of Region of interest: Using the coordinates the License plate area is cropped and extracted

3) Preprocessing the ROI(Region of Interest): Various image transformations are performed in general to enhance the image quality. The ROI is subjected to Gray Scaling, Gaussian Blur, and Denoising to recognize the characters easily.

4) *Character Recognition:* After the image preprocessing characters from the number plate are extracted using one of the cost-effective Optical Character Recognition tools named EasyOCR. The last step is to check whether the Extracted License plate is in the CSV file and produce the result as College bus if it is present and NON-COLLEGE BUS else. Figure 1 shows the architecture diagram of college bus number plate registration Using YOLO V8.

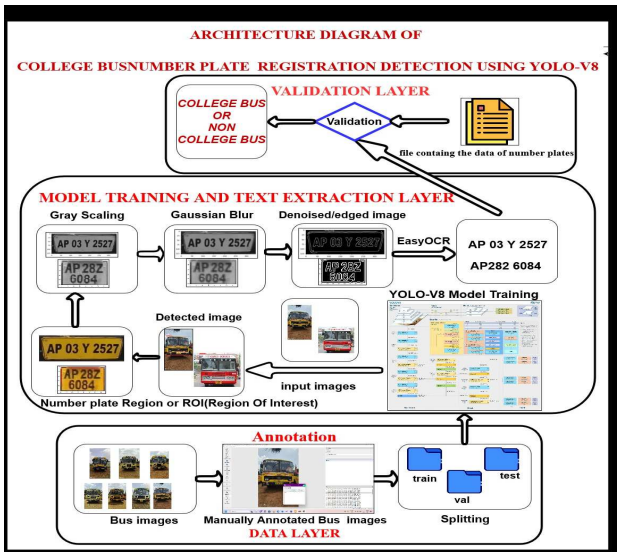

Fig. 1. Architecture diagram for college bus number plate Registration Using YOLO V8

VII. METHODS

A. Data Annotation

A dataset of BVRIT Buses is collected and annotation (Drawing the bounding boxes) is performed and labels are given as: 1. Bus 2. Number plate. The labels are saved in YOLO format. The images and labels are Split into Train and test data. In Fig. 2, it shows the image before detection.

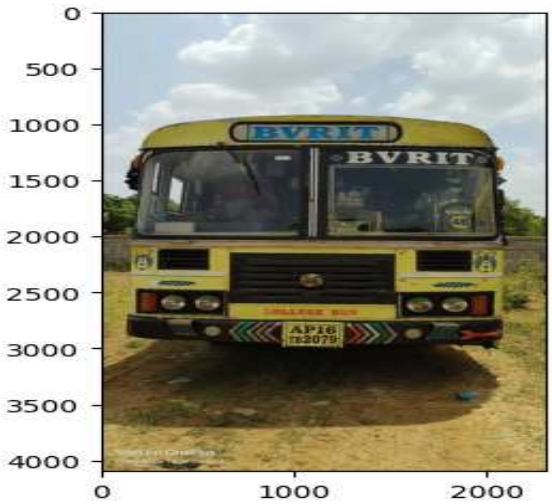

Fig. 2. Bus image before the detection

B. Training the YOLO v8 model

Create a file having filename "custom. yaml" for custom object detection. Yolo v8 model is trained for 50 epochs for accurate detection of buses and number plates. Fig. 3, shows Bus image after detection. The model is subjected to prediction to test model. We obtain the detected image with two classes detected: bus and number plate. In Fig. 3, extracted number plate region is shown.

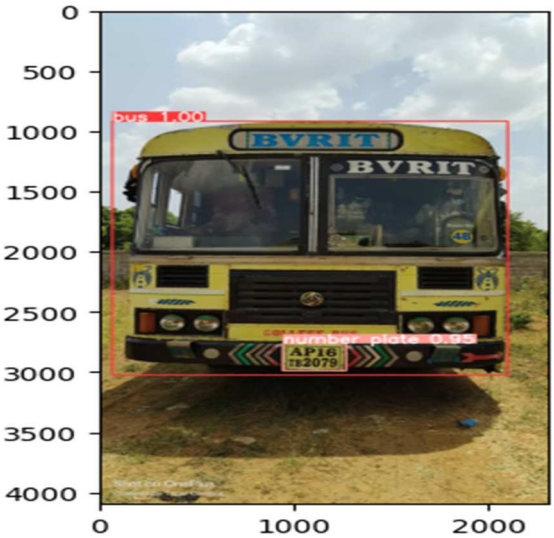

Fig. 3. Bus image after the detection using YOLO v8

The region of interest is extracted from the coordinates of bounding boxes of the number plate obtained from the detection. In Fig.4.,It shows the extracted ROI or Number plate from the detection.

Fig. 4. ROI(Region of Interest) or Number plate region

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C. Gray Scale

Gray Scale conversion is very useful for the recognition and analysis of images, as it is an important image preprocessing technique It requires maintaining essential visual characteristics while transforming a color image into gradations of gray. By excluding color information and focusing solely on intensity variations, grayscale graphics simplify the display of visual data. Gray scaling of an image is calculated using (5):

$$gray\ scaling = \frac{R+G+B}{3} \tag{5}$$

R=RED, G=GREEN, B=BLUE

D. Gaussian Blur

A typical method for smoothing images is Gaussian blur, it is one of the image preprocessing steps in character recognition to reduce noise and increase the legibility of the characters.The following equation (6) can be used to add Gaussian blur to an image:

$$I\ (x,y) = \frac{1}{2\pi\sigma^2}\sum_{neighborhood_pixels} e^{-\left(\frac{(x-\mu)^2+(y-v)^2}{2\sigma^2}\right)} \tag{6}$$

E. Character Recognition using "Easy OCR"

After the detection is performed and ROI is Extracted and preprocessed the characters are extracted using EasyOCR. A popular open-source optical character recognition (OCR) software called EasyOCR offers a straightforward and practical method for extracting text from. It supports more than 80 languages and is built on top of the PyTorch deep learning platform. EasyOCR uses preprocessing methods to improve recognition precision. In order to make the text easier to read, it automatically rotates the images, modifies the contrast, and applies adaptive thresholding.

The output obtained is:['AP167B2079']

VIII. Formulas

A. Precision

Precision measures the proportion of true positive predictions out of all positive predictions made by a model. It is calculated using equation (7):

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \tag{7}$$

B. Recall

Recall, also known as sensitivity or true positive rate, measures the proportion of true positive predictions out of all actual positive instances in the dataset. It is calculated using equation (9)

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{8}$$

C. mAP-50

mAP-50 is a performance metric commonly used in object detection and recognition tasks. It calculates the average precision at a specific intersection over union (IoU) threshold, usually set to 0.5. It is calculated using equation (9):

$$mAP_{50} = \frac{AP_{0.5}+AP_{0.55}+AP_{0.6}\cdots+AP_{0.95}}{N} \tag{9}$$

where AP at 0.5, AP at 0.55..., AP at 0.95 represent the average precision at different IoU thresholds, and N represents the Number of IoU thresholds used.

D. mAP-(50 to 95)

mAP at 50-95 is similar to mAP at 50 but calculates the average precision in a range of IoU thresholds from 0.5 to 0.95. It is calculated using equation (10):

$$mAP_{50} = \frac{AP_{0.5}+AP_{0.55}+AP_{0.6}\cdots+AP_{0.95}}{M} \tag{10}$$

where AP at 0.5, AP at 0.55..., AP at 0.95 represent the average precision at different IoU thresholds from 0.5 to 0.95, and M is the total number of IoU thresholds in that range.

IX. Results

The proposed College bus Number plate Registration detection model using YOLO V8 uses a three-module flow to recognize the College bus or NON- COLLEGE BUS. At very fast rate, the image is captured and sent as an input to the custom Yolo V8 model for accurate detection of the Bus and its Number plate with a high confidence value. The Model accurately detected the bus and Number plate using the Single Shot detection technique for which YOLO is the best choice.After the detection the Bounding Boxes of the number plate are extracted using the inbuilt functions provided by YOLO V8 as it is an anchor-free detection and which R-CNN, OpenCV2, and previous models could not give. After the extraction of the Bounding box of the Number plate, the coordinates of the bounding Boxes are extracted and in turn, the Region of Interest is obtained from the image, just using the coordinates of the Bounding box, all other previous models and existing system use the contours for the extraction of the Region of Interest (ROI). The coordinates are extracted accurately using the YOLO v8. This makes YOLO V8 Unique from the other existing systems.After the Extraction of the ROI, image processing techniques like GrayScaling, Gaussian Blur, and Denoising are performed, and Edge detection is also performed on the ROI to improve the accuracy of the recognition of the text. to perform the recognition techniques like EasyOCR as it uses automatic Adaptive Thresholding on the image and extracts the text from it. The text is extracted with a minimum number of errors. This extracted text is now checked in the CSV file containing the data of the License plate of the buses and if it is present we get the output as College bus or else NON-COLLEGE BUS. We took two test images(1 belongs to bvrit and 1 Non-college bus) and performed the experiment the results obtained are :

1. The two buses and number plate are detected accurately using the YOLO V8 model.
2. The coordinates of the Number plate region are extracted from the detected buses.
3. The text from the number plates is extracted using EasyOCR for the two Buses.
4. The extracted text is checked in the CSV file.
5. For the Bus which belongs to College(BVRIT) the output obtained as College bus and for NON-College bus we got output as NON-COLLEGE BUS.

Below table 1 shows the results obtained by YOLO V8:

TABLE I. Results of overall college bus number plate using Yolo v8

Results Using YOLO V8				
Class	Precision	Recall	mAP50	mAP50at95
Overall	99.2%	95%	99%	83.7%
Bus	98.8%	100%	99.5%	94%

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Number plate	99.5%	90%	98.6%	73.4%

TABLE II. OVERALL COMPARISON OF EXISTING MODELS YOLO V5,YOLO V7,YOLO V8.

Comparison Along with Previous Models				
Versions	Precision	Recall	mAP50	mAP50at95
YOLO V5	0.90	0.85	0.92	0.88
YOLO v7	0.88	0.82	0.90	0.85
YOLO V8	99.2%	95%	99%	83.7%

From the above table 2 we can conclude that YOLO V8 is better model for detection. In Fig.5 The F1 score curve, which is obtained after training the YOLO V8 model is show.

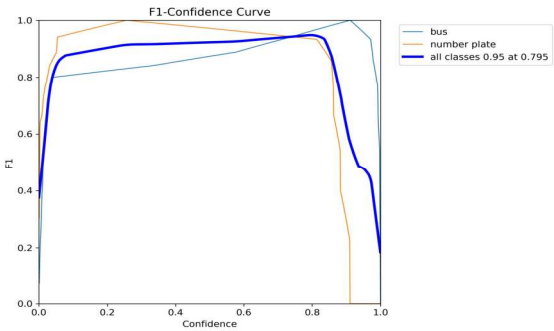

Fig. 5. F1 curve

In Fig.6 The Precision score curve, which is obtained after training the custom YOLO V8 algorithm is shown.

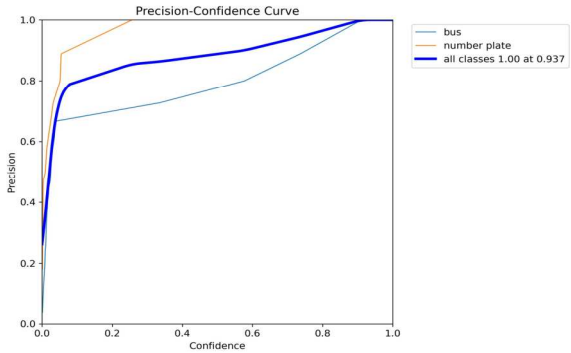

Fig. 6. Precision curve

In Fig.7 The Confusion matrix, which is obtained after training the YOLO V8 model is shown.

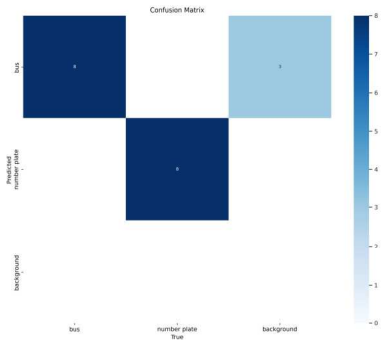

Fig. 7. Confusion matrix

The confusion matrix is a table used to evaluate the performance of the YOLO v8 model or any model. It compares the predictions made by the model with the actual labels and gives information about predictions, true positives (TP)(which are predicted accurately by the model for which it is trained), true negatives (TN)(which are over-predicted by the model), false positives (FP)(which are predicted incorrectly, that is a different entity is predicted instead it is not trained for detecting that entity by the model), and false negatives (FN) (Incorrectly predicting the entities for which it is trained for). In Fig.8 The Confusion matrix, which is obtained after training the YOLO V8 model is shown.

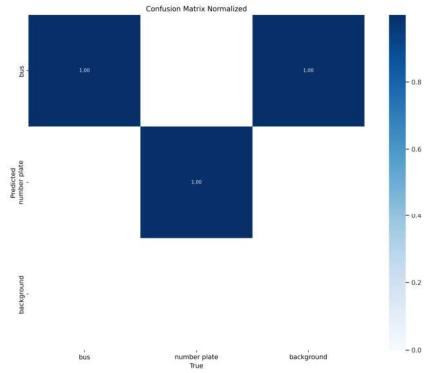

Fig. 8. Normalized Confusion matrix

Normalized confusion matrix is obtained from the confusion matrix by dividing it with the total number of instances the training is done. System Requirements include Hardware Requirements are GPU(Graphical Processing unit) RAM to train model and Software Requirements are Google Collab for executing, YOLO v8 is repository is required. Decent Internet connection is required.

X. DISCUSSION

Detection of Bus and number plate can be done using many approaches like using deep Learning, OpenCV, Machine Learning algorithms, and YOLO. YOLO V8 algorithm jhas used in this research, YOLO v8 is latest version of acclaimed real-time object detection and image segmentation model. YOLO-v8 is built on cutting-edge advancements in deep learning and computer vision, offering unparalleled performance in terms of speed and accuracy. Its streamlined design makes it suitable for various applications and easily adaptable to different hardware platforms, from edge devices to cloud APIs. YOLO v8 uses anchor-free detection for detecting objects. AS YOLO v8 belongs to its Single Shot Detection family it can detect many objects in one single forward pass, whereas Deep Learning and OpenCV are Two Shot Detection families, they take more time compared to YOLO. Another important feature of YOLO v8 is, we can directly extract Bounding Box from detected image without performing the contours and extracting longest contours. As YOLO is trained on COCO dataset, it can detect objects with high accuracy and in any other weather conditions. YOLO v8 can detect vehicles in low-light conditions also. YOLO v8 research was primarily motivated by empirical evaluation of COCO benchmark. As each piece of network and training routine are tweaked, new experiments are run to validate the change's effect on COCO modeling. Training of YOLOv8 will be probably faster than other two-stage object detection models. At current time, YOLOv8 does not support models trained in 1280 (in pixels) resolution. Advantage of anchor-free detection is we don't need to manually specify the anchor boxes, which is not present in previous YOLO versions 1 and 2. The only correction for proposed model is, Model cannot

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detect numbers when numbers on number plate are eroded due to paint as we are using EasyOCR, It can directly Recognize the Characters which are similar to the English language. By comparison with other existing models proposed model can detect number plate and bus in less time and with high accuracy.

XI. Conclusion

Research implements College bus and Number Plate registration detection using YOLO V8 in three Modulus for detecting whether bus is College bus or NON-COLLEGE BUS, first module focus on Bus and Number plate detection using YOLO v8 model trained on customed dataset, the second module focuses on the Extraction of Number plate using the bounding boxes obtained from YOLO V8 detected results, the third module focuses on recognition of text from number plate. Necessary preprocessing techniques are performed in Second module to enhance recognition of characters using EasyOCR. Research combines Object detection framework and Optical Character Recognition framework(EasyOCR) for recognition of License plate. In conclusion, the model's overall performance is remarkable. While maintaining a recall of 94.2 percent, which emphasizes its capacity to collect a substantial number of pertinent things, it attained a precision of 95.7 percent, suggesting a high accuracy in properly identifying objects. Model's ability to detect items with a high degree of confidence is demonstrated by its mAP50 score of 99.5 percent. Additionally, its efficiency in recognizing objects over larger range of confidence levels is demonstrated by the mAP50-95 score of 86.0 percent. The "bus" class showed a precision of 91.5 percent when analyzing specific classes, demonstrating a great capacity to precisely distinguish buses among other things. Model's 100 percent recall means that it was able to identify every instance of the "bus" class. The model's capacity to recognize buses with a high degree of certainty is further supported by the mAP50 score of 99.5 percent, and the mAP50-95 score of 94.8 percent shows that it is effective at doing so across a range of confidence levels. Model received a perfect precision score of 100 percent for the "number plate" class, proving its accuracy in identifying number plates. Even though the recall rate of 88.3 percent indicates that some number plates may have been overlooked, model nevertheless managed to catch a sizable portion of them. The model's proficiency at identifying number plates across a wider range of confidence levels is highlighted by its mAP50-95 score of 77.3 percent, which is higher than its mAP50 score of 99.5 percent. Overall, the model's high object detection abilities are demonstrated by these performance indicators, especially in the "bus" and "number plate" classes. The model's ability to precisely detect and localize objects in images is supported by its high precision and recall values and impressive mAP scores. This is one of optimal solutions to conserve time and for smart planning and transport management of the organization. Limitations are ,due to the variability of number plate formats might struggle in detection and conventional formats like different regions having different designs, It may give false positive FP (detecting non-number plate objects like number plates) or else false negative FN(missing actual number plates Future scope of this model may get improved in dynamic context through their potential for real-time adjustment to change conditions. Accuracy of model can improve over a range of circumstance thru cooperative efforts to produce number plate board and training dataset.

Acknowledgment

This Research study is supported by Ms. Ayesha Naureen Assistant Professor, M. Tech (Ph.D.). We thank our mentor Ms. Ayesha Naureen Assistant Professor, M. Tech (Ph.D.), who contributed greatly to our research study and instructed the insights which assisted the research, from our team members.

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